1 ISOLATED POWER FROM EXTERNAL 5 V

VBUS DOMAIN / GND_ISO

VBus powers only the RX divider.
U1 powers the isolated RX/TX interface,
independently of bus levels and pauses.
R1: rail discharge and reference load.
Do not connect GND to GND_ISO.

Resistors: 1%, 0805. Bypass: 100n, 50V X7R.
C1/C3/C13: 22u, 25V X7R, 1210.
C5: 22u, 25V X7R, 1210.

MCU DOMAIN / GND

U2
MIC5235-3.3YM5-TR
+3V3_ISO
R1 3k00
C5 22u / 25V
VISO_RAW
C3 22u / 25V
C4 100n
GND_ISO

U1
R05CTE05S-CT
VOUT+ VOUT+ VOUT- VOUT- SGND_OUT SGND_OUT SGND_OUT DNC
VIN+ VIN+ VIN- VIN- SGND_IN SGND_IN SGND_IN SGND_IN
GND
+5V
C1 22u / 25V
C2 100n
C13 22u / 25V
J1 EXTERNAL 5 V

C8 100n
C6 100n
C7 100n
+3V3_ISO
+3V3_MCU

2 POLARITY-INDEPENDENT VBUS / RX WITH HYSTERESIS

J2 VBUS
VBUS_A
VBUS_B
D1 SMBJ12CA
D2 BAT54S
D3 BAT54S
VBUS_RECT
R2 25k5
R4 330k
R3 10k
C10 220p C0G 50V
SENSE
RX_REF
U3 TS391RIYLT
IN+ IN- V+ V- OUT
RX_ISO
R12 10k
GND_ISO

U4 ISO7221ADR
INB_(RX) OUTA_(TX) INA_(TX) OUTB_(RX)
VCC1 VCC2 GND1 GND2
TX_ISO
R7 100R
RX_MCU
TX_MCU
R8 10k
+3V3_MCU
GND_ISO
GND

U7 D1 mini / S3
3V3 5V
D7_/_S3:11 D6_/_S3:13
RST_/_EN A0_/_S3:2 D0_/_S3:4 D5_/_S3:12 D8_/_S3:10 TX_/_S3:43 RX_/_S3:44 D1_/_S3:36 D2_/_S3:35 D3_/_S3:18 D4_/_S3:16
GND

Nominal VBUS_RECT thresholds: about 2.01 / 2.26 V.
C10 filters noise. U3 uses the ST TS391R pin assignment.

U4: ISO7221ADR, both sides at 3.3 V.
Do not fit a default-LOW isolator.

3 TX WITH UNDERVOLTAGE INHIBIT

U6
SN74AUP1G97DBVR
+3V3_ISO
GND_ISO
R9 100k
U5 MAX809TEUR+T
VDD RESETn GND
ISO_READY
C11 100n
C12 100n
TX_ISO
C_/_TX B_/_READY A_=_0 VCC Y GND
R10 100R
Q1 NTR4003NT3G
TX_GATE
R11 10k
VBUS_RECT
R5 43k
R6 10k
RX_REF
C9 4n7 C0G 100V

TP1 VISO_RAW
TP2 +3V3_ISO
TP3 GND_ISO
TP4 RX_ISO
TP5 TX_GATE
TP6 VBUS_RECT
TP7 +5V
TP8 +3V3_MCU
TP9 GND

U6: Y = READY AND NOT TX.
TX HIGH: bus released. TX LOW: Q1 on.
U5: 3.08 V threshold; release delay 140..560 ms.

UART 9600 8N1, non-inverted.
D1: RX13 / TX12. S3: RX11 / TX13.
Disconnect external 5 V before USB.
TX requires firmware bus arbitration.

Jacopo Labardi / @labodj / personal use
Sheet: /
File: Resol_VBus_RevE.kicad_sch
Title: RESOL VBus — Rev E / Isolated RX + TX
Size: A3 | Date: 2026-10-18 | **Rev: E**
KiCad E.D.A. 10.0.6 | Id: 1/1